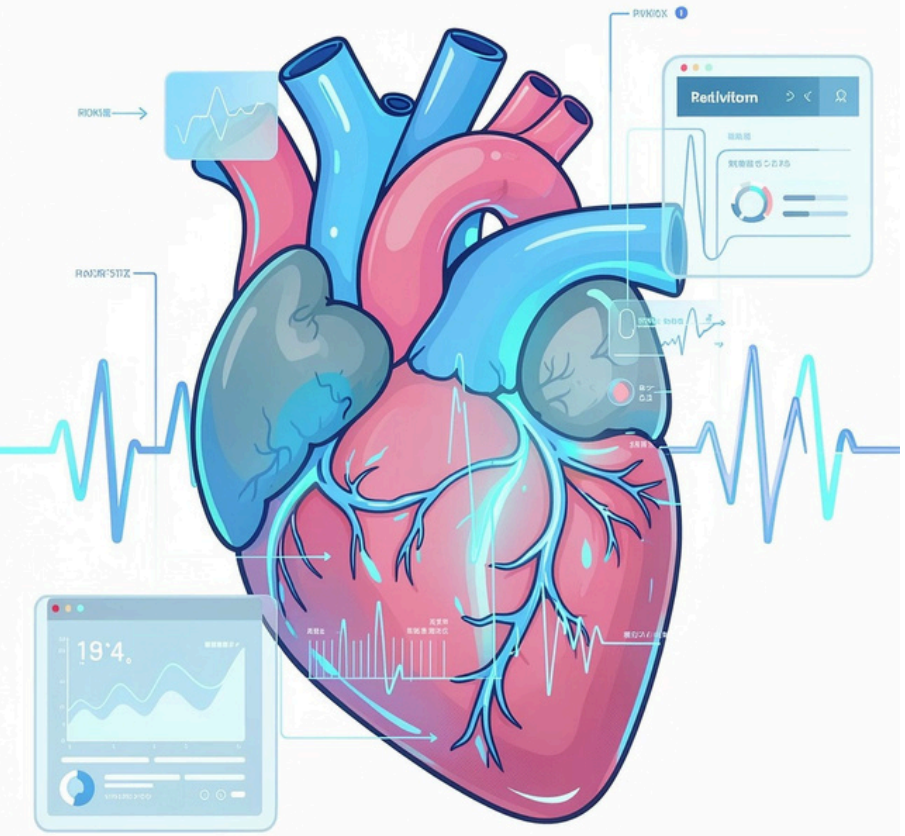

Heart Disease Prediction using Machine Learning

Create a machine learning model to forecast heart disease by utilizing clinical data.

Presented By:

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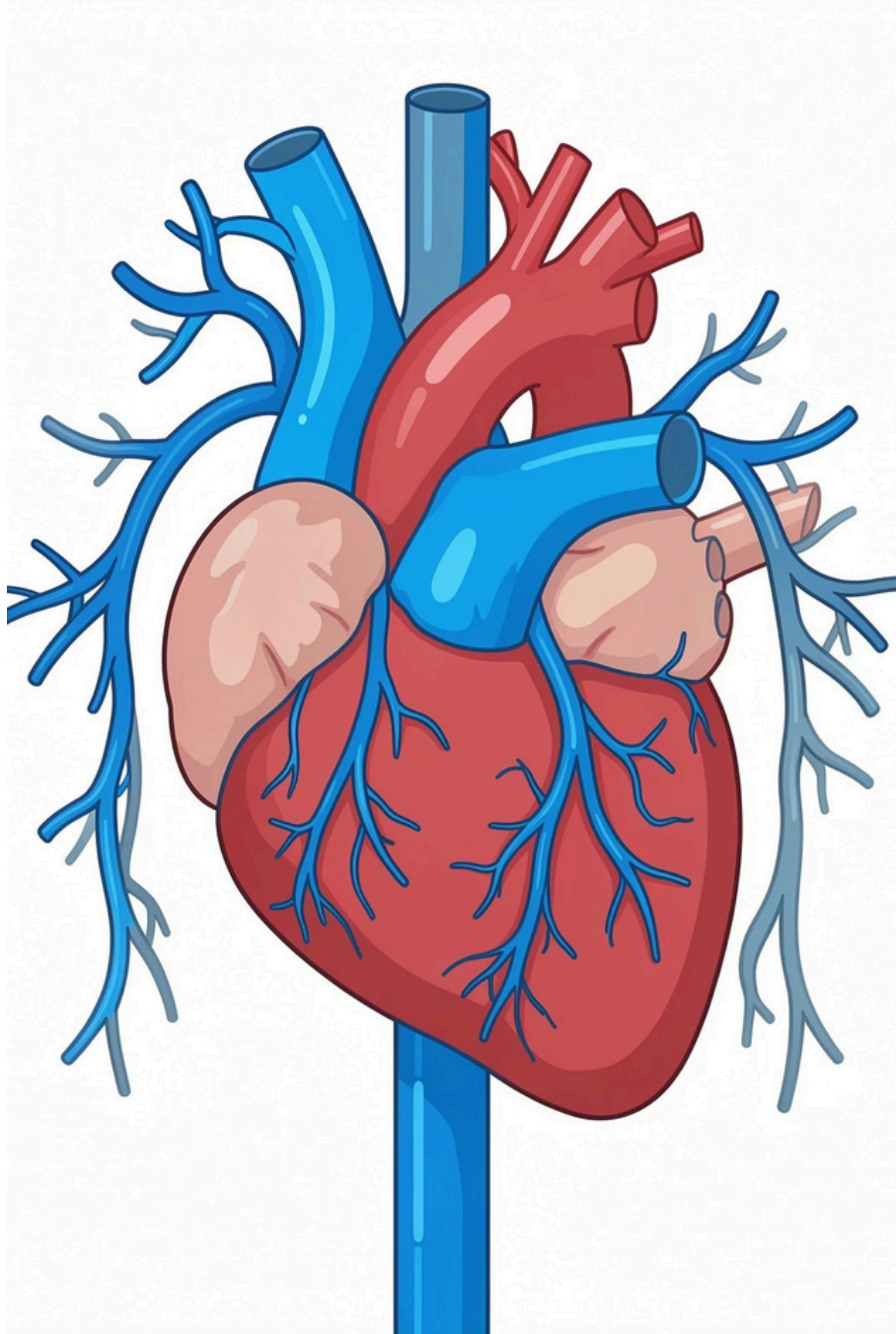
The Challenge: Cardiovascular Disease

The Global Burden

Cardiovascular diseases (CVDs) continue to be the foremost cause of death globally, responsible for around 17.9 million fatalities each year, according to data from the World Health Organization (WHO).

Our Mission

Develop a machine learning model that predicts heart disease presence with high confidence using clinical data, enabling early intervention and improved patient outcomes.



Cleveland Dataset: Foundation of Our Model

303

Patient Records

Comprehensive clinical datafrom the UCI Machine Learning Repository

13

Features

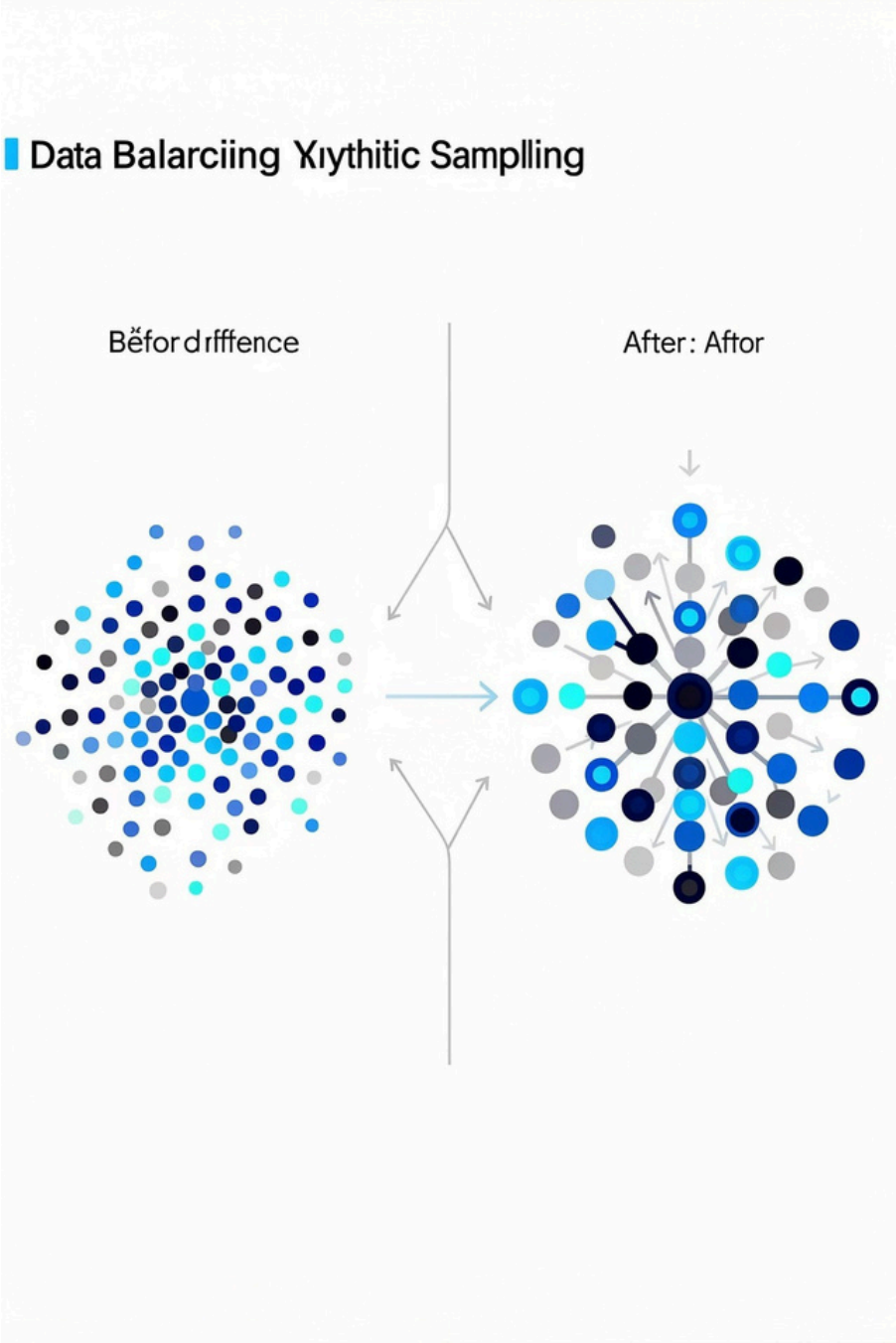
Clinical indicators including age, sex, chest pain type, blood pressure, cholesterol levels, and maximum heart rate

2

Classes

Binary classification: healthy patients (0) versus heart disease patients (1)

Data sourced from Cleveland Clinic Foundation, providing a well-established benchmark for cardiac diagnostic research.



Addressing Data Imbalance with SMOTE

1

The Problem

Original datasetshowed class imbalance with more healthy patients than diseased cases, risking model bias

2

The Solution

Applied SMOTE(Synthetic Minority Over-sampling Technique) to generate synthetic minority class samples

3

The Outcome

Preventedclassifier bias and significantly improved recall performance for disease detection

Model Selection Process

1

Logistic Regression

Baselinelinearmodelfor comparison

2

K-Nearest Neighbors

Non-parametricapproach based on similarity

3

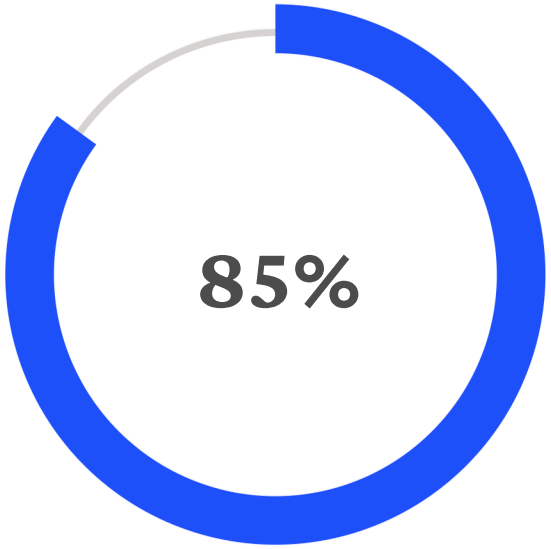
Gradient Boosting

Selectedmodel:handles non-linear relationships effectively

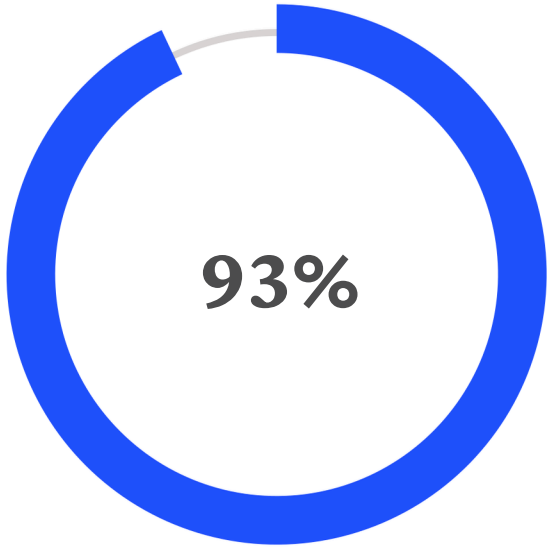
Why Gradient Boosting?

- Excels at capturingcomplex, non-linear patterns in medical data
- Builds decision trees sequentially to correct previous errors
- Provides robust handling of feature interactions

Performance Metrics: The Proof



Accuracy
Overall correctness across both classes



ROC-AUC Score
Outstanding discriminatory ability

ROC-AUC Interpretation: An AUC of 0.93 means the model has a 93% probability of correctly distinguishing between a patient with heart disease and a healthy patient—a clinically significant diagnostic capability.

Feature Importance: What Drives Predictions

Chest Pain Type (cp)

Strongest clinical indicator for heart disease presence

Number of Vessels (ca)

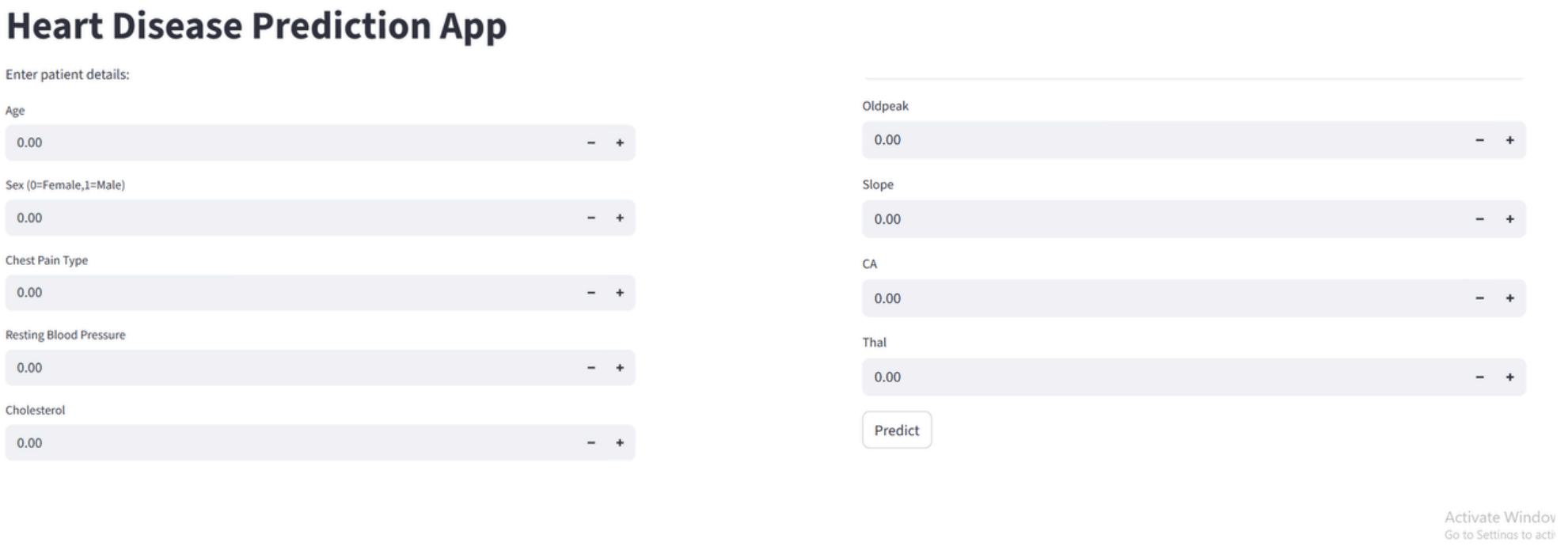
Coronary artery visibility directly impacts diagnosis

Maximum Heart Rate (thalach)

Achieved heart rate during stress testing

These clinically interpretable features enable doctors to understand model decisions, building trust in AI-assisted diagnosis while maintaining transparency.

Live Deployment: Streamlit Application



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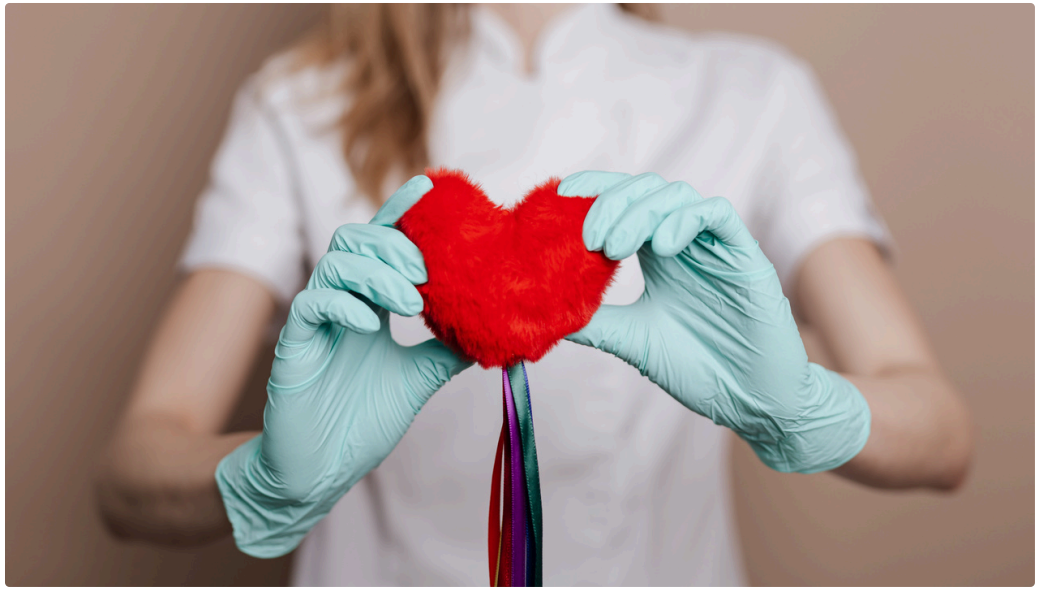

FromResearchtoClinical Tool

DeployedonStreamlitCloudforreal-time predictions with user-friendlyinterface.

Key Features

- Interactive sliders for clinical parameters
- Instant prediction with confidence level
- Probability percentages (80%, 90%, etc.)
- Mobile-responsive design

Impact: This model serves as a decision-support tool for clinicians to prioritize high-risk patients and optimize resource allocation in cardiac care.